

PDU Firmware – Proposed API, CAN & MQTT Data Flow

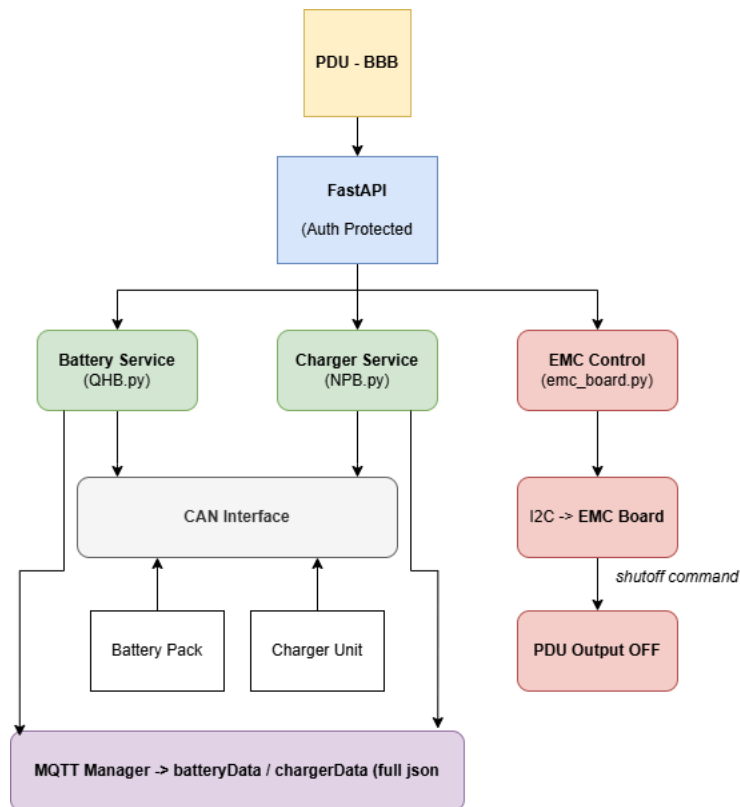
1. Purpose

The PDU firmware monitors Battery and Charger data continuously over CAN, decodes it, and publishes the full telemetry to an MQTT broker in real time. On top of this, FastAPI exposes a small, authenticated control surface limited to three functions for this phase:

- Configuring where the MQTT data is published (broker/topics).
- A lightweight status check for quick health/at-a-glance monitoring.
- A remote shutdown command routed to the EMC board over I2C.

Detailed, continuously-changing battery and charger data is not exposed through repeated API polling — that data already flows through MQTT. The API layer stays deliberately thin.

2. Architecture Diagram



3. Flow

3.1 Data Collection & Publishing (continuous)

Battery CAN → qhb.py ↴

—▶ Latest Data Object

Charger CAN → NPB.py ↴

—▶ MQTT Manager —▶ batteryData / chargerData (full JSON)

—▶ FastAPI —▶ GET /status (summary subset only)

3.2 Remote Shutdown (on demand)

Client → FastAPI Auth → Remote Shutdown API → emc_board.py → I2C → EMC Board → PDU outputs OFF

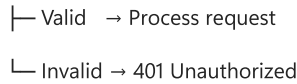
4. Authentication

FastAPI uses existing username/password authentication. For this scope, authentication protects:

- MQTT configuration (GET/POST /api/v1/mqtt/config)
- Remote shutdown (POST /api/v1/control/remote shutdown)

The status endpoint is read-only telemetry and does not require authentication, unless the team decides otherwise.

Client → FastAPI Authentication



5. MQTT Configuration API

Configures the broker connection used for continuous publishing. Once configured, the Battery and Charger services publish their full decoded JSON payloads (the same structures already produced by qhb.py and NPB.py) to their respective topics — the API itself does not carry that data.

	Endpoint	Method	Auth	Description
1	/api/v1/mqtt/config	GET	Yes	Returns current broker host, port, and topic names.
2	/api/v1/mqtt/config	POST	Yes	Updates broker host, port, and topic names.

POST /api/v1/mqtt/config — request body

```
1 {
2   "broker": "192.168.1.100",
3   "port": 1883,
4   "broker_username": "ADC",
5   "broker_password": "df"
6   "battery_topic": "PDU/<hostname>/batteryData",
7 }
```

</> JSON

Once saved, MQTT publishes continue automatically — the full battery and charger JSON objects are pushed to these topics on every update, without further API calls.

6. Status API

A single lightweight endpoint for a quick health snapshot. It intentionally returns only the essentials — not the full decoded payload — since detailed data is already streaming continuously through MQTT.

	Endpoint	Method	Auth	Description
1	/api/v1/status	GET	No*	Returns a minimal battery/system summary.

*Can be switched to authenticated if required.

Example response

</> JSON

```
1 {
2   "hostname": "pdu-bbb-001",
3   "mac_address": "XX:XX:XX:XX:XX:XX",
4   "timestamp": "2026-08-10T13:20:30Z",
5   "battery_percentage": 82,
6   "battery_state": "Discharging",
7   "system_voltage": 51.8,
8   "load_current": 4.2,
9   "battery_charging_current": 11,
10  "battery_capacity_in_min": 120,
11  "time_since_boot_in_sec": 2340
12 }
```

	≡ Field	≡ Meaning
1	battery_percentage	Battery state of charge (SoC), %
2	power source	battery / AC in
3	system_voltage	Present DC bus voltage (battery pack voltage)
4	load_current	Present current draw on the DC bus
5	battery charging current	+ve for charging current, - ve for discharging
6	battery capacity in minutes	amount of time left
7	time since boot in secs	

7. Remote Shutdown API

Triggers a physical power shutdown of the PDU outputs via the EMC board over I2C. This is a hardware action, distinct from stopping the monitoring software.

	≡ Endpoint	≡ Method	≡ Auth	≡ Description
1	/api/v1/control/remote shutdown	POST	Yes	Sends a shutdown command to the EMC board.

POST /api/v1/control/remote-shutdown — request body

```
{
  "turn_pdu-_off_remotely": True
}
```

	≡ command	≡ Action
1	0	Turn PDU outputs OFF
2	User name	user who send the command
3	Date and time stamp	Timestamp (UTC format)
4		

Additional command values (e.g. remote power-on) are not defined in this scope and would need confirmation before implementation.

Response

</> JSON

```
1 {
2   "success": true,
3 }
```